

Network 1

Sections 1:3

Made By Ibrahim Selim

Section 1: Installing Programs and Introduction

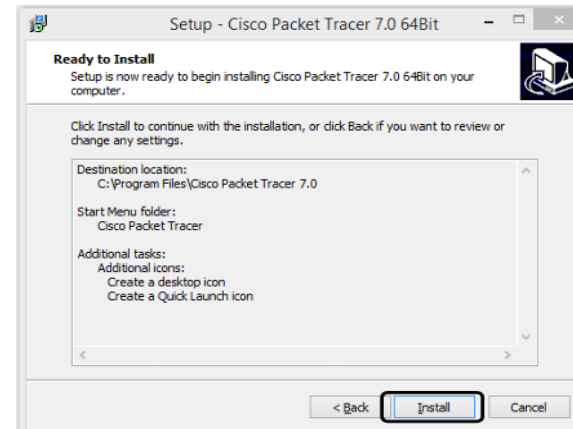
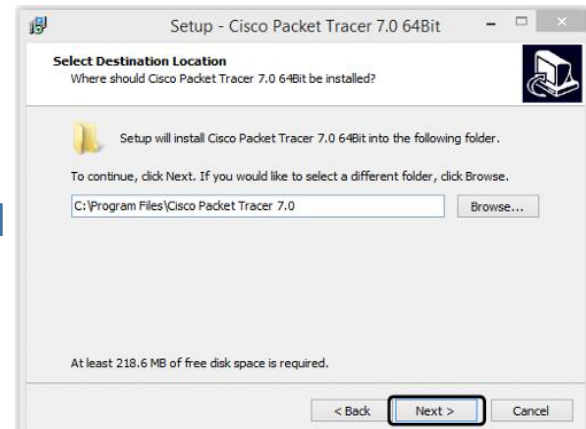
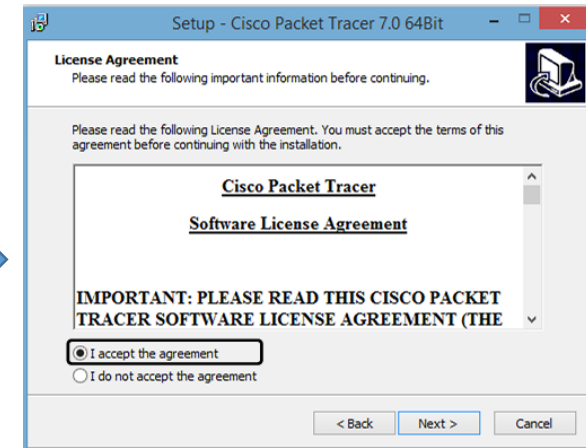
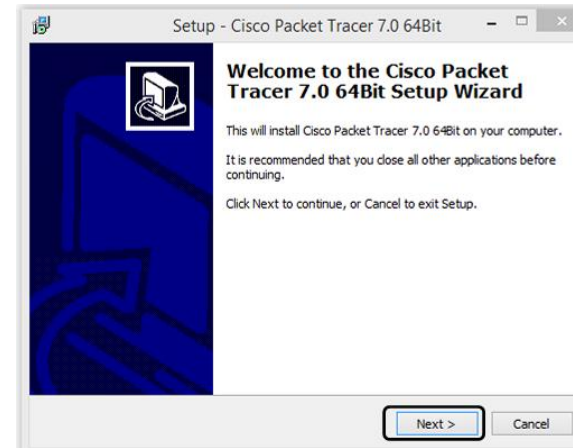
Cisco Packet Tracer

- Instead of using real routers and switches (which are expensive), Packet Tracer lets you simulate a network on your computer.
- You can:
 - Add routers
 - Add switches
 - Connect PCs
 - Configure IP addresses
 - Test routing protocols
 - Simulate DHCP, DNS, HTTP, etc.

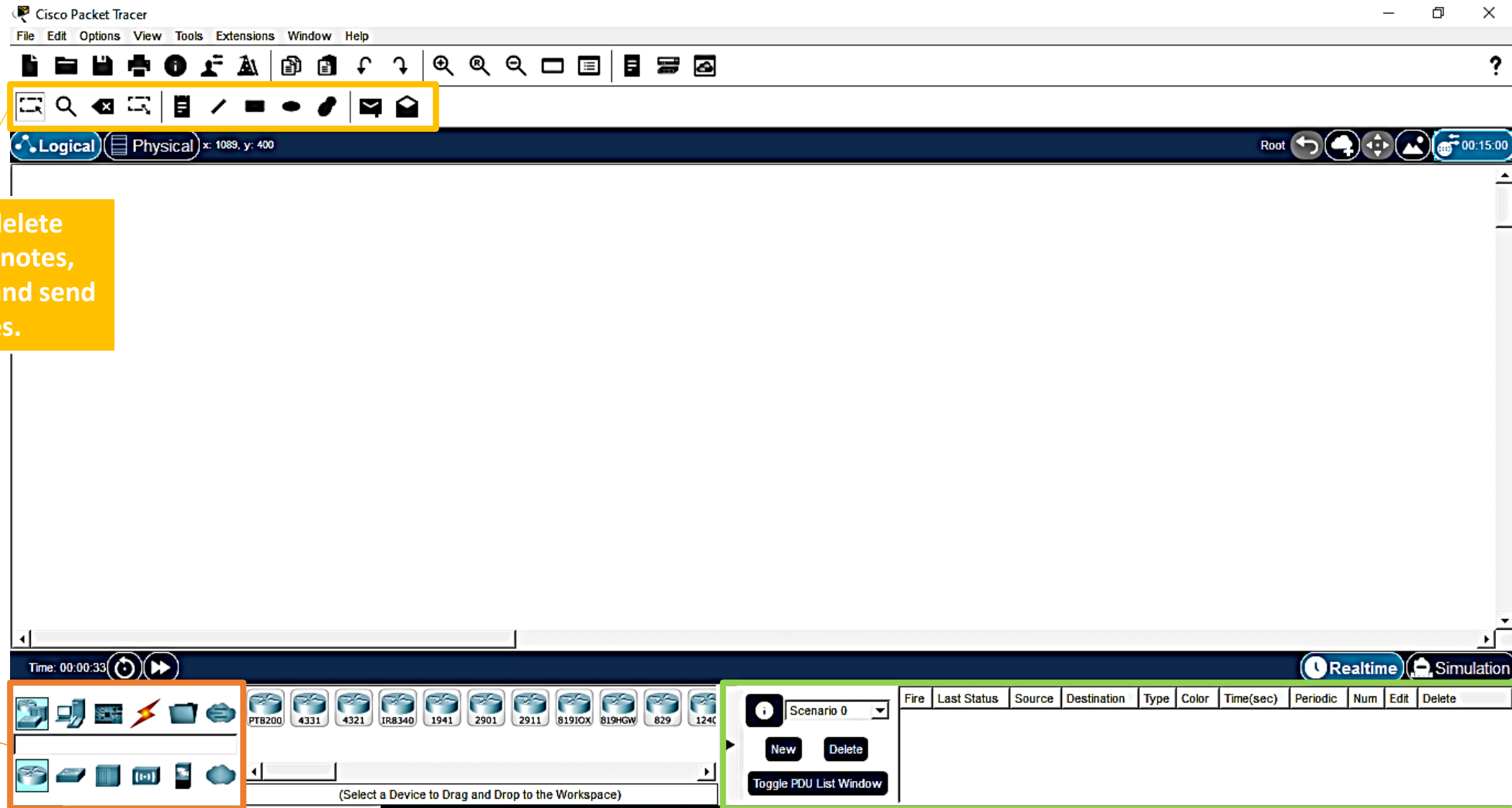


Cisco Packet Tracer installation

- 1) Create a Free Cisco Account
<https://www.netacad.com/>
- 2) Go to Resources → Download Packet Tracer
 - Choose your operating system:
 - Windows 64-bit or 32-bit
 - Linux
 - macOS
- 3) Open the downloaded file
- 4) Next → Accept → Next → Install



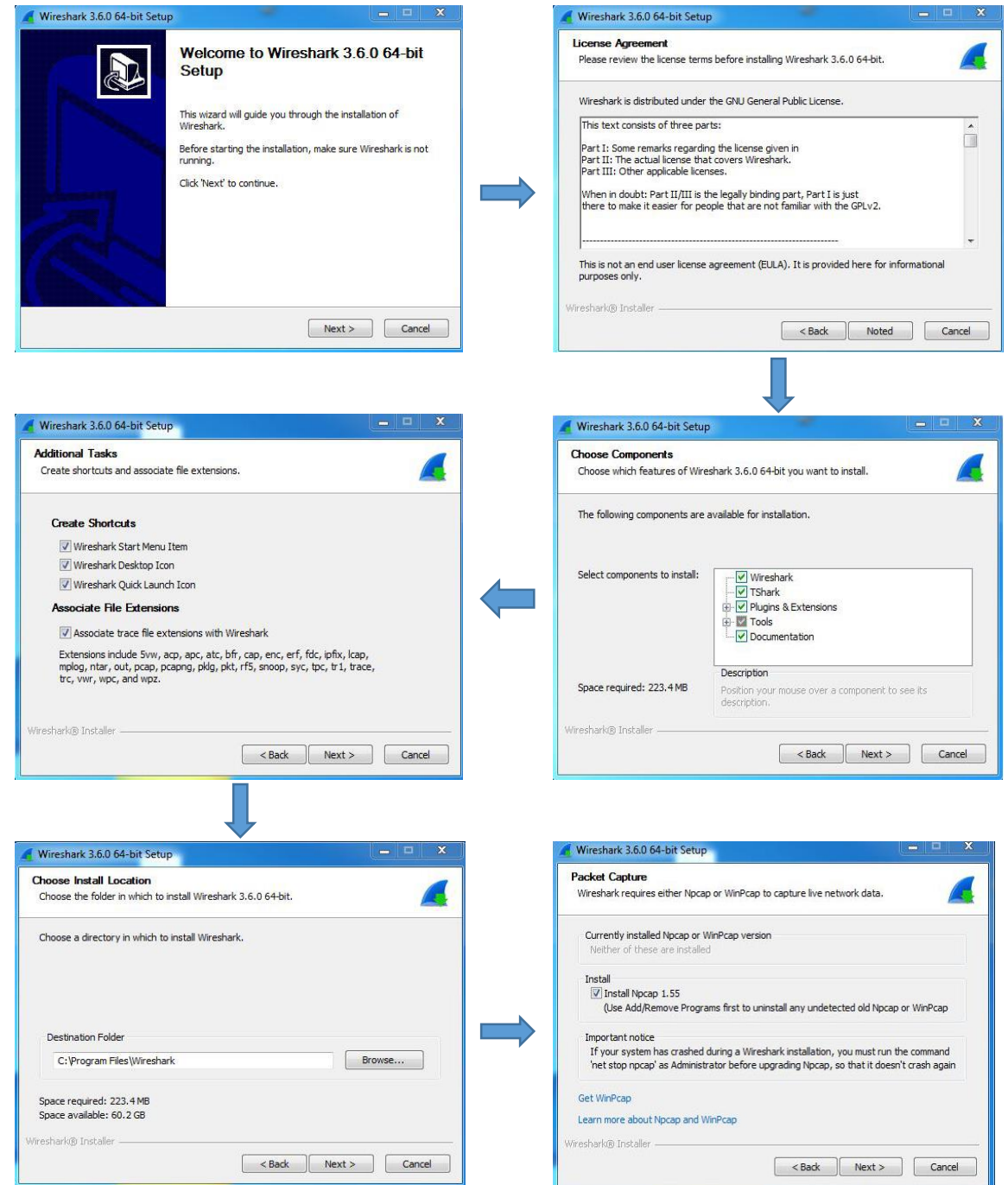
Cisco Packet Tracer



Wireshark



- Wireshark is a free network protocol analyzer used to capture and analyze network traffic (packets) on your computer
- Download Wireshark
 - Go to the official website: <https://www.wireshark.org>
 - Click Download.
 - Choose Windows Installer (64-bit).
- Run the Installer:
 - During installation, Wireshark will ask to install Npcap.
 - Next → Next → Next → Install → Finish



Wireshark

The image shows the Wireshark network protocol analyzer interface. The top pane, 'Packet List', displays a table of captured packets. The middle pane, 'Packet Details', shows the hierarchical structure of the selected packet (No. 349). The bottom pane, 'Packet Bytes', shows the raw data of the selected packet in a hexdump format.

Packet List Pane: Displays all the packets in the current capture file. The table shows columns for No., Time, Source, Destination, Protocol, Length, and Info.

No.	Time	Source	Destination	Protocol	Length	Info
343	65.142415	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519346 TSecr=551811827
344	65.142715	192.168.0.21	174.129.249.228	HTTP	253	GET /clients/netflix/flash/application.swf?flash_version=flash_lite_2.1&v=1.5&n
345	65.230738	174.129.249.228	192.168.0.21	TCP	66	80 → 40555 [ACK] Seq=1 Ack=188 Win=6864 Len=0 TSval=551811850 TSecr=491519347
346	65.240742	174.129.249.228	192.168.0.21	HTTP	828	HTTP/1.1 302 Moved Temporarily
347	65.241592	192.168.0.21	174.129.249.228	TCP	66	40555 → 80 [ACK] Seq=188 Ack=763 Win=7424 Len=0 TSval=491519446 TSecr=551811852
348	65.242532	192.168.0.21	192.168.0.1	DNS	77	Standard query 0x2188 A cdn-0.nflximg.com
349	65.276870	192.168.0.1	192.168.0.21	DNS	489	Standard query response 0x2188 A cdn-0.nflximg.com CNAME images.netflix.com.edge
350	65.277992	192.168.0.21	63.80.242.48	TCP	74	37063 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=491519482 TSecr
351	65.297757	63.80.242.48	192.168.0.21	TCP	74	80 → 37063 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM=1 TSval=3295
352	65.298396	192.168.0.21	63.80.242.48	TCP	66	37063 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=491519502 TSecr=3295534130
353	65.298687	192.168.0.21	63.80.242.48	HTTP	153	GET /us/nrd/clients/flash/814540.bun HTTP/1.1
354	65.318730	63.80.242.48	192.168.0.21	TCP	66	80 → 37063 [ACK] Seq=1 Ack=88 Win=5792 Len=0 TSval=3295534151 TSecr=491519503
355	65.321733	63.80.242.48	192.168.0.21	TCP	1514	[TCP segment of a reassembled PDU]

Packet Details Pane: Shows the current packet (selected in the "Packet List" pane) in a more detailed form. The details are expanded for the Domain Name System (response) section.

```
> Frame 349: 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits)
> Ethernet II, Src: Globalsc_00:3b:0a (f0:ad:4e:00:3b:0a), Dst: Vizio_14:8a:e1 (00:19:9d:14:8a:e1)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.0.21
> User Datagram Protocol, Src Port: 53 (53), Dst Port: 34036 (34036)
  > Domain Name System (response)
    [Request In: 348]
    [Time: 0.034338000 seconds]
    Transaction ID: 0x2188
    > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 4
    Authority RRs: 9
    Additional RRs: 9
  > Queries
    > cdn-0.nflximg.com: type A, class IN
  > Answers
  > Authoritative nameservers
```

Packet Bytes Pane: Shows the data of the current packet (selected in the "Packet List" pane) in a hexdump style.

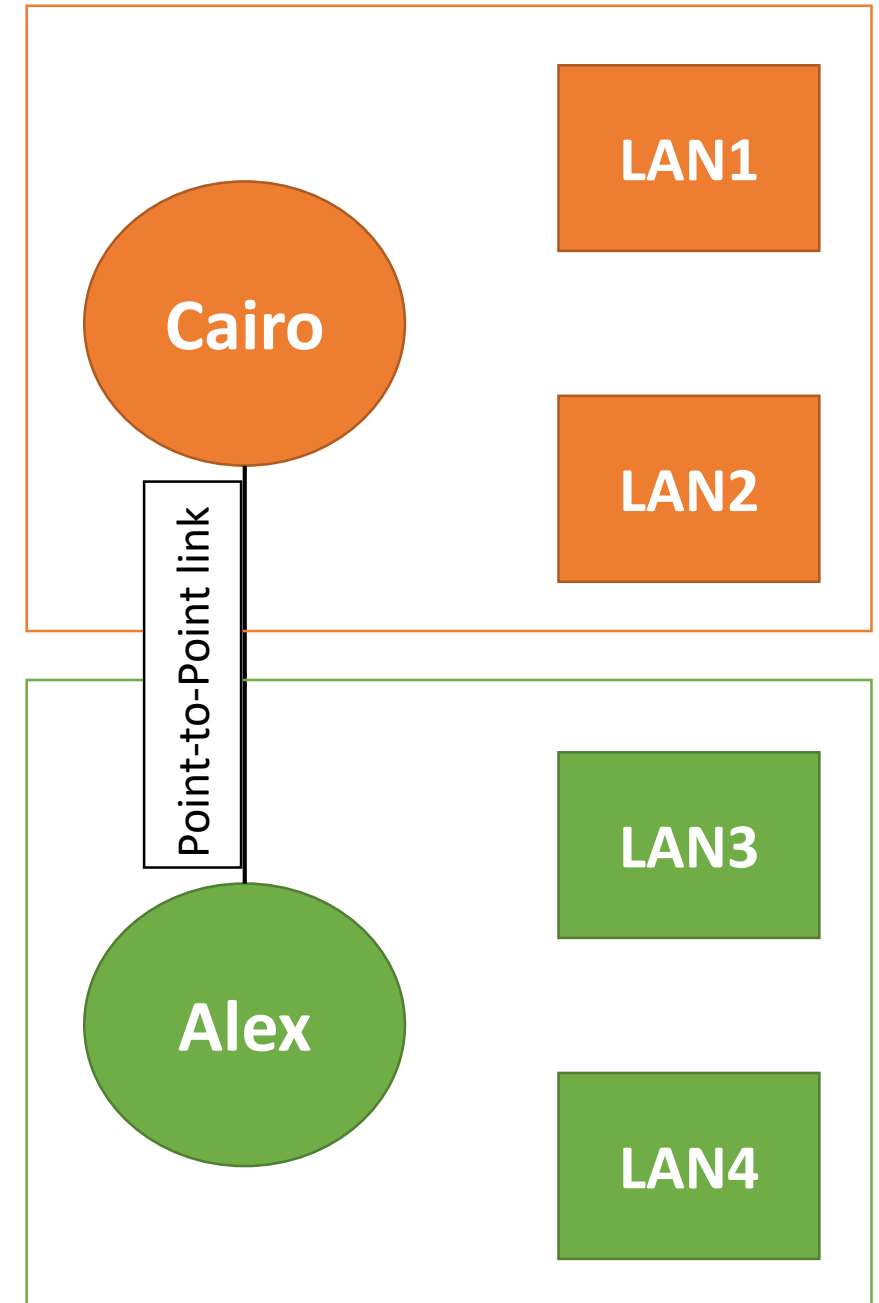
```
0020  00 15 00 35 84 f4 01 c7 83 3f 21 88 81 80 00 01 ...5....?l....
0030  00 04 00 09 00 09 05 63 64 6e 2d 30 07 6e 66 6c .....c dn-0.nfl
0040  78 69 6d 67 03 63 6f 6d 00 00 01 00 01 c0 0c 00 ximg.com .....
0050  05 00 01 00 00 05 29 00 22 06 69 6d 61 67 65 73 .....). ".images
0060  07 6e 65 74 66 6c 69 78 03 63 6f 6d 09 65 64 67 .netflix .com.edg
0070  65 73 75 69 74 65 03 6e 65 74 00 c0 2f 00 05 00 esuite.n et../...
```

Section 2: Static VLSM Subnetting

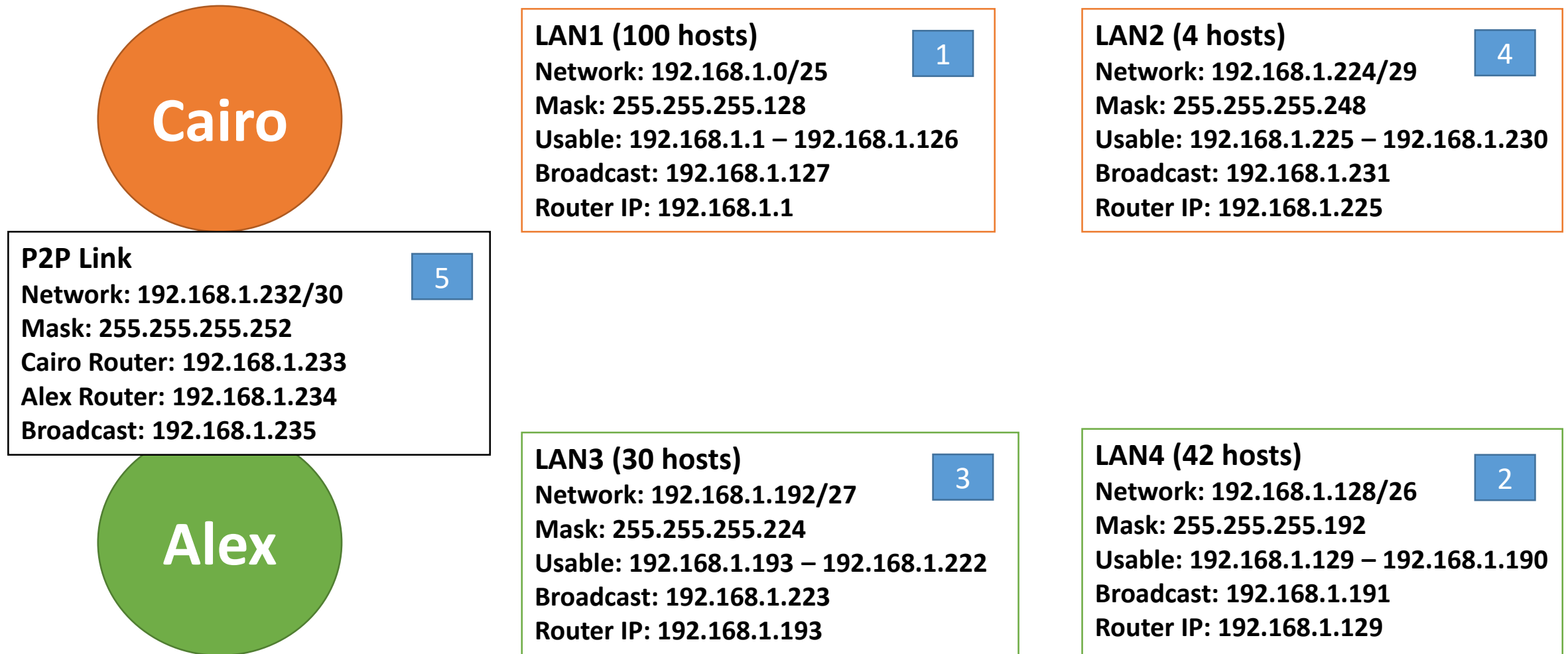
Static VLSM Subnetting

- Variable Length Subnet Masking (VLSM) is an efficient IP addressing technique that allows for creating subnets of different sizes, preventing address waste.
- Unlike FLSM (Fixed Length), VLSM minimizes wasted IP addresses, crucial for IPv4 conservation.
- Example: Company has 2 branches, Cairo has two LANs and Alex has two LANs and P2P between Cairo and Alex.
- You have a single network: 192.168.1.0/24
 - (256 addresses total)
- You need to subnet it Starting from 192.168.1.0 using VLSM for:
 - LAN 1 → 100 hosts (Cairo)
 - LAN 2 → 4 hosts (Cairo)
 - LAN 3 → 30 hosts (Alex)
 - LAN 4 → 42 hosts (Alex)
 - Point-to-Point link between Cairo & Alex routers → 2 hosts

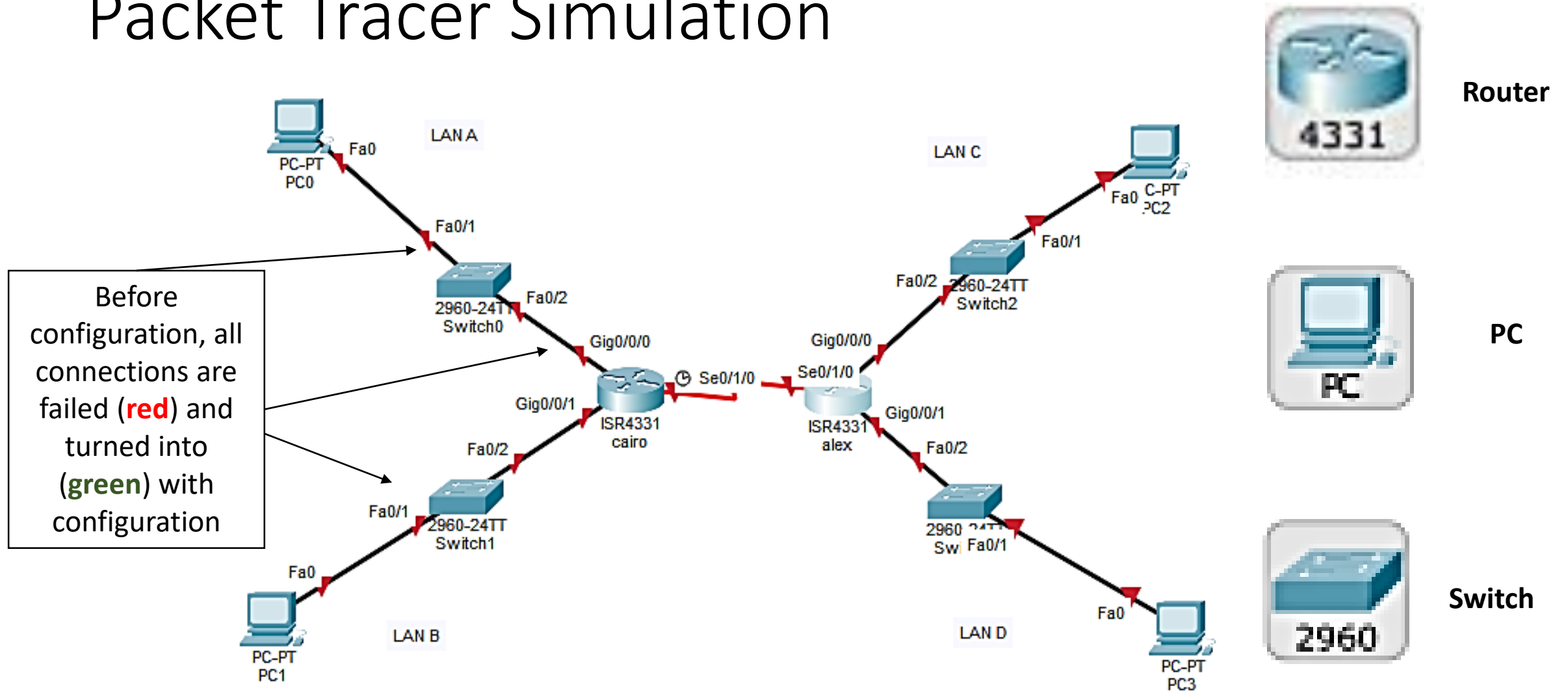
$$2^n - 2 \geq \text{hosts}$$



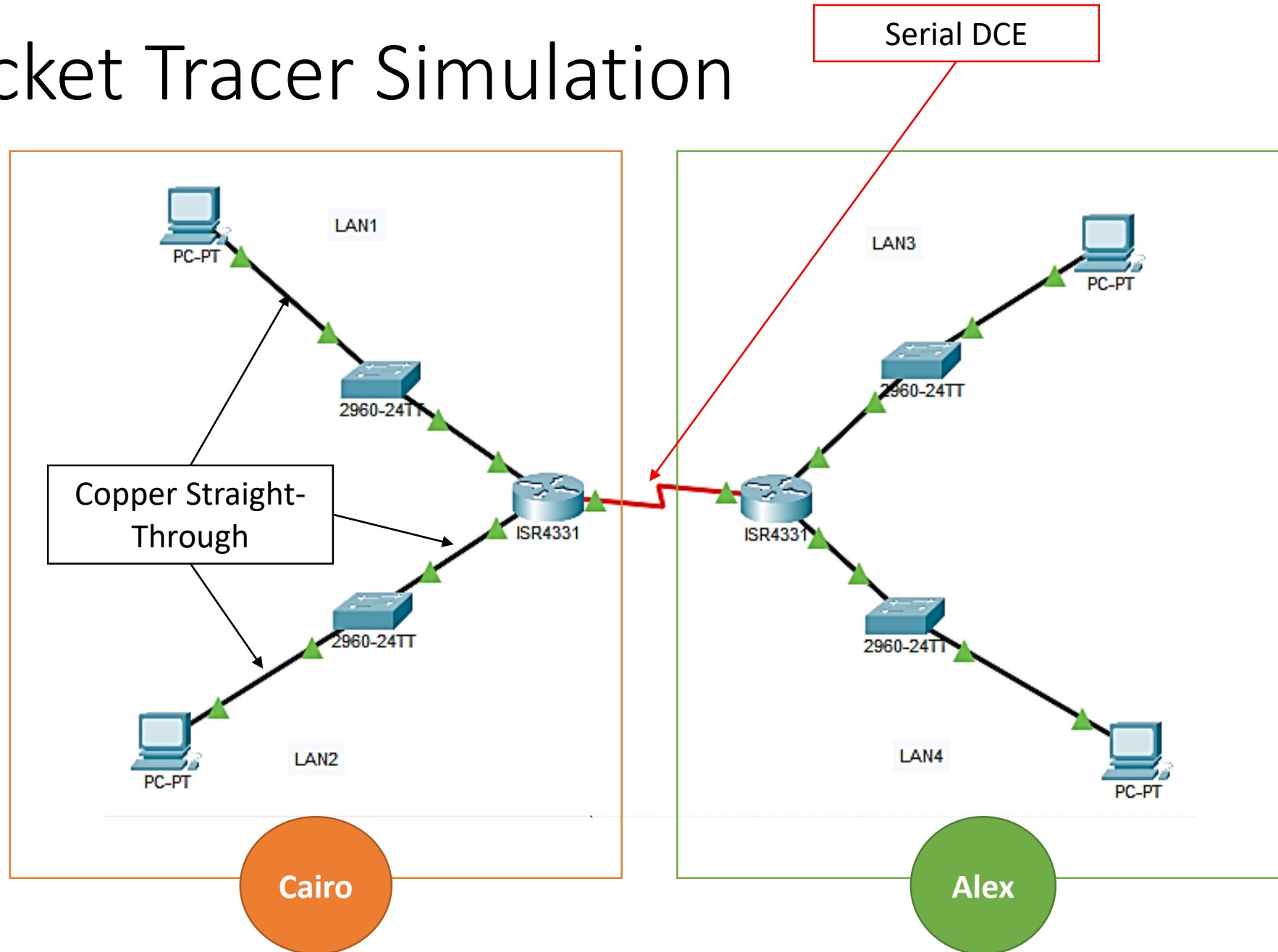
Subnet Allocation (Largest to Smallest)



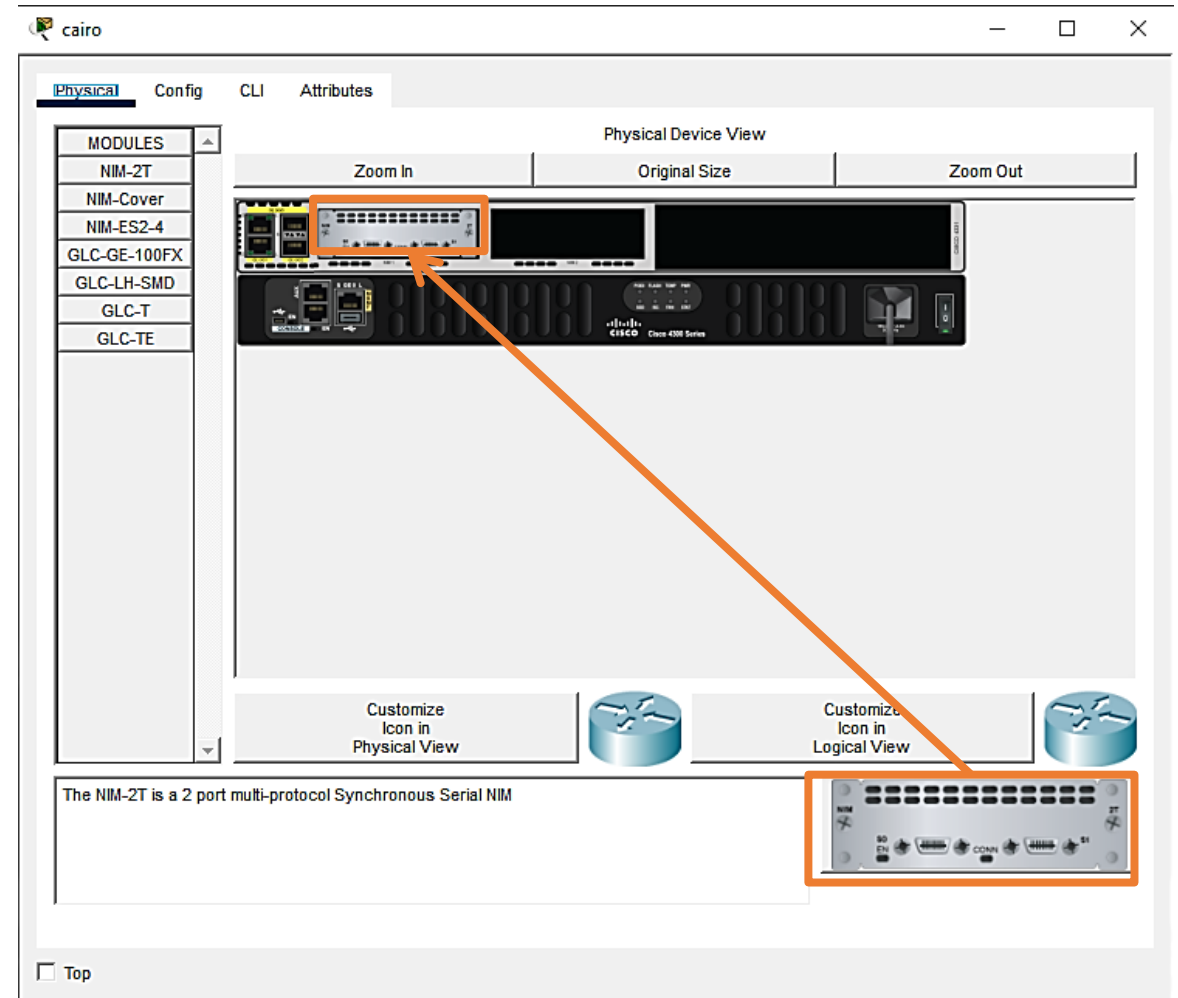
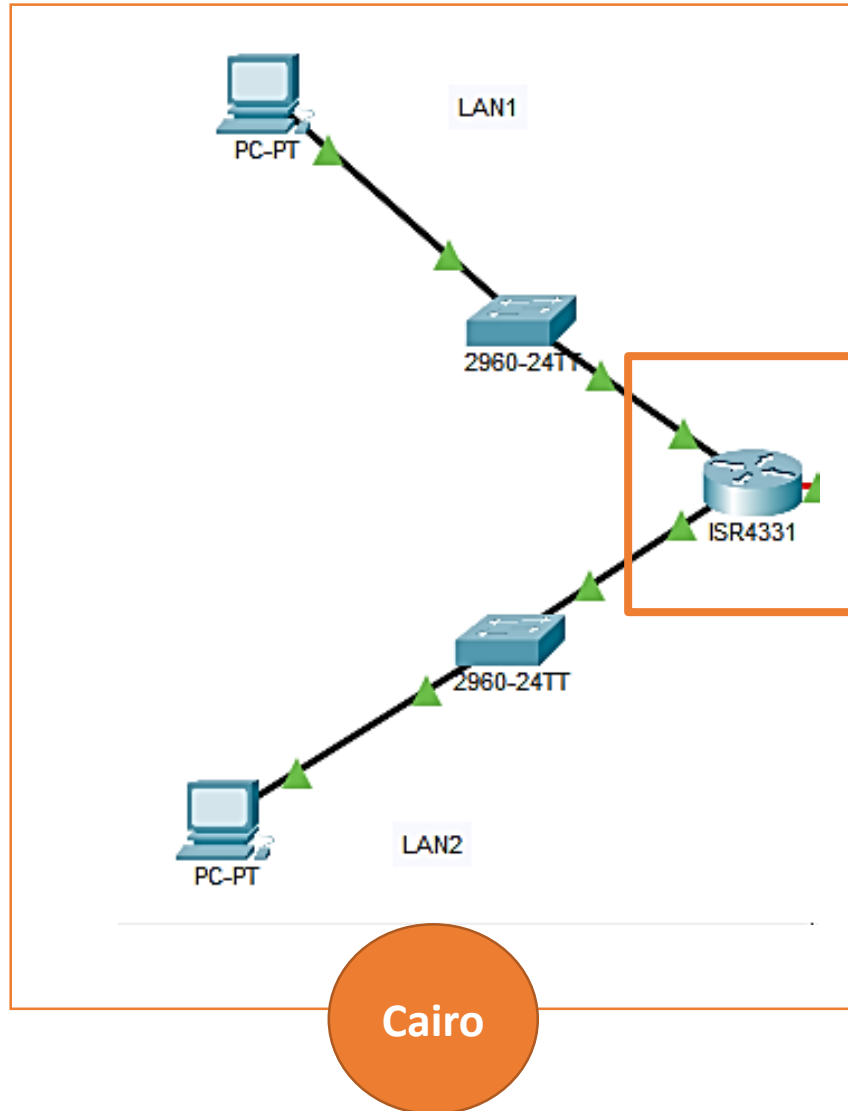
Packet Tracer Simulation



Packet Tracer Simulation



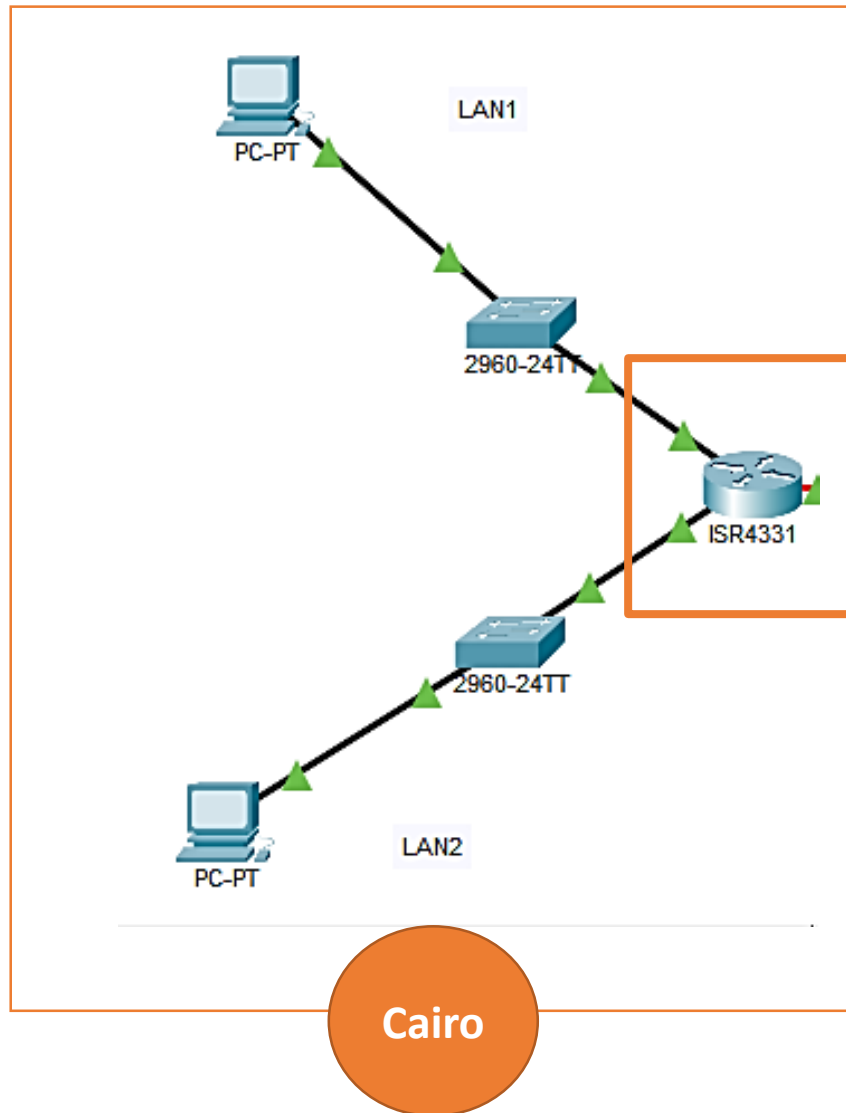
Static Routing (Packet Tracer)



Add serial Serial NIM interface to each routers.
Note: turn off routers before adding

Static Routing (Packet Tracer)

Add IP and Subnet Mask to each interface and turn on port status.



cairo

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0/0
- GigabitEthernet0/0/1
- GigabitEthernet0/0/2
- Serial0/1/0
- Serial0/1/1

GigabitEthernet0/0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.5818.3101

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.128

Tx Ring Limit 10

Equivalent IOS Commands

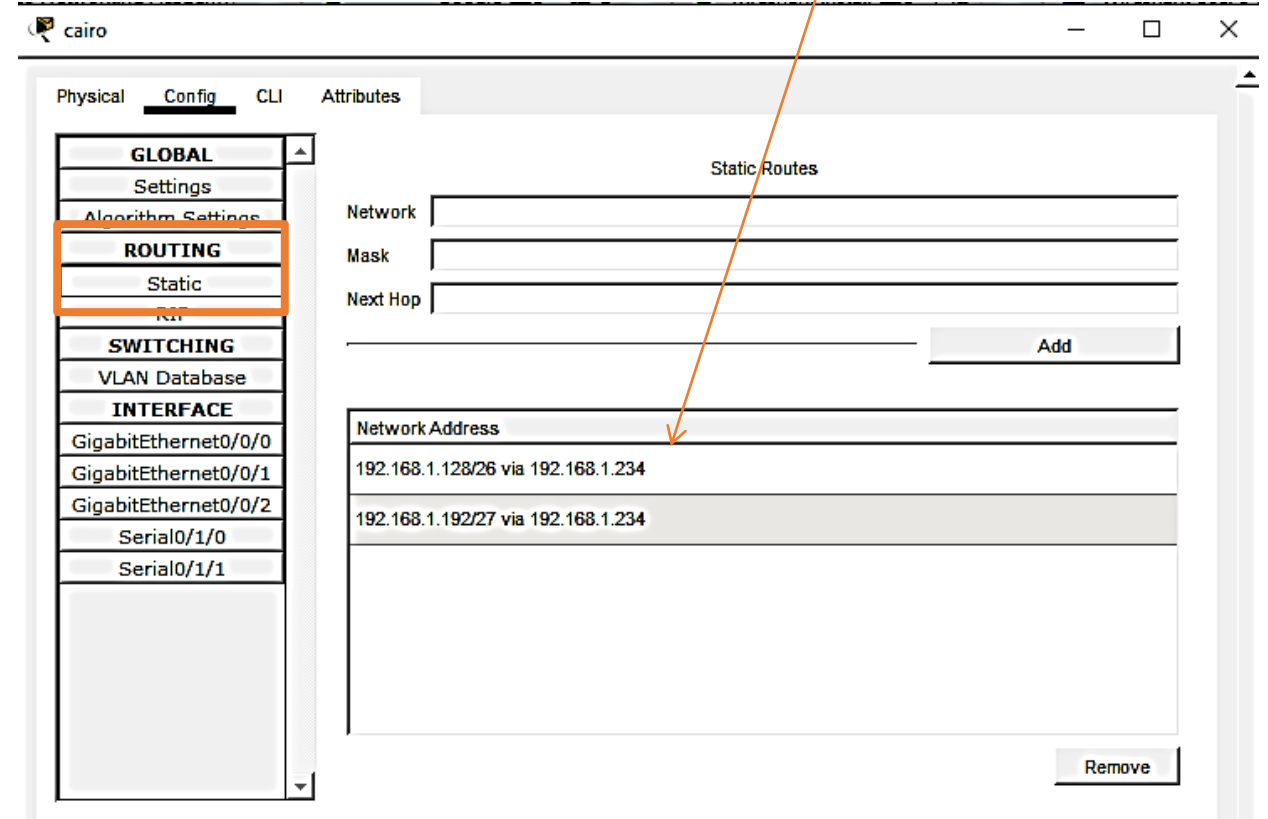
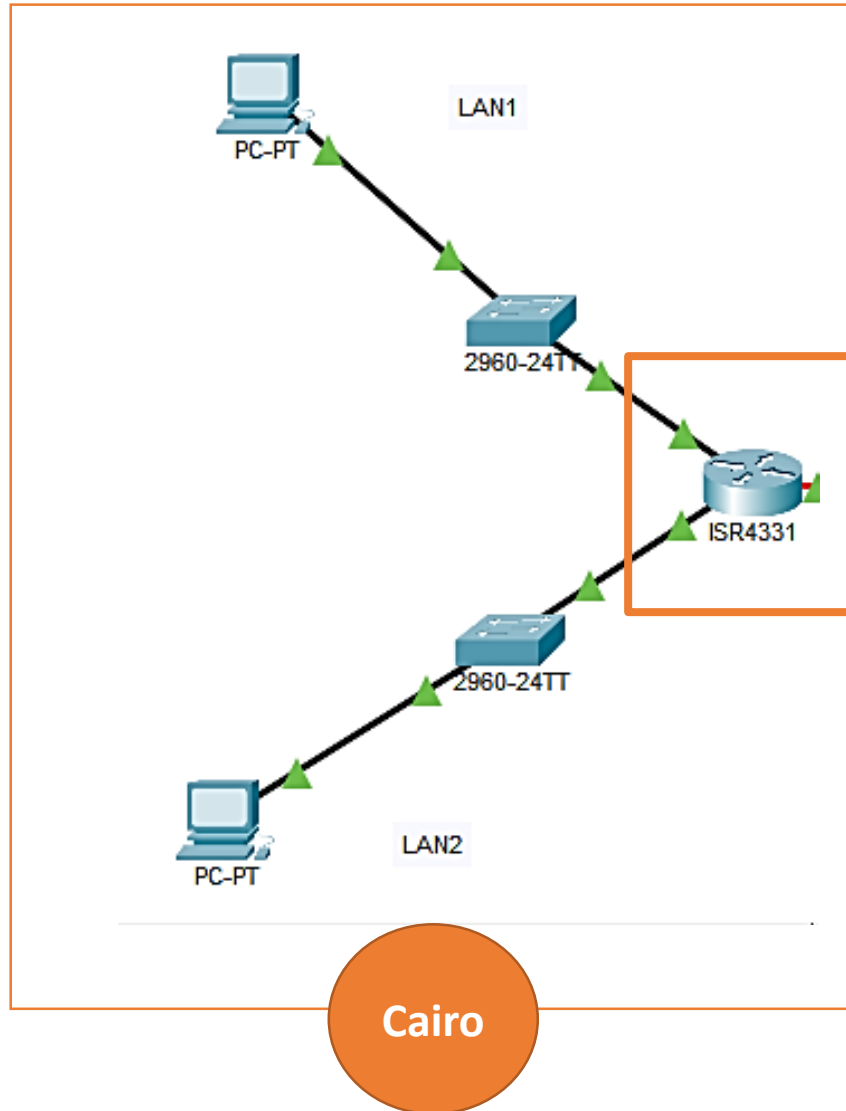
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
```

☐ Top

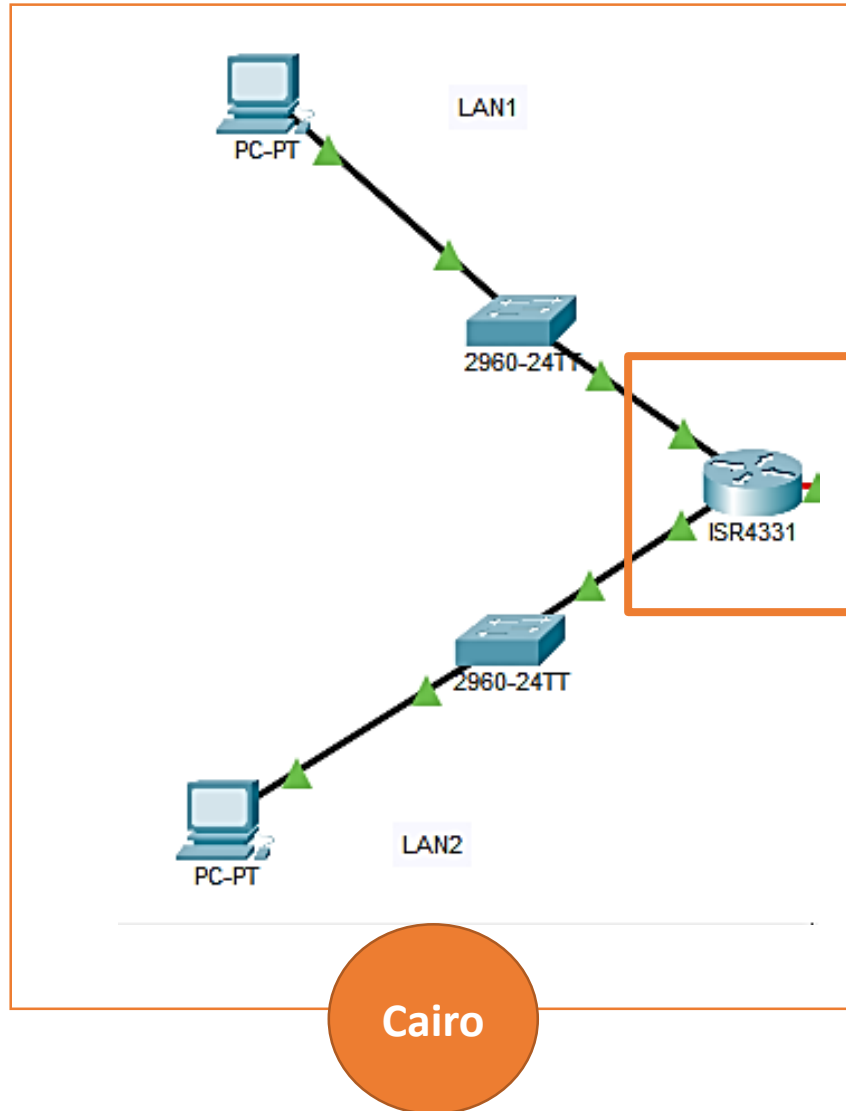
Static Routing (Packet Tracer)

Add Static Routing to each router to see other networks in another router.



Static Routing (Packet Tracer)

You can add IP and Subnet Mask to each interface, turn on port status and add static routing to each router to see other networks in another router using CLI commands.



```
cairo
Physical Config CLI Attributes
IOS Command Line Interface

enable
conf t

interface g0/0/0
ip address 192.168.1.1 255.255.255.128
no shutdown

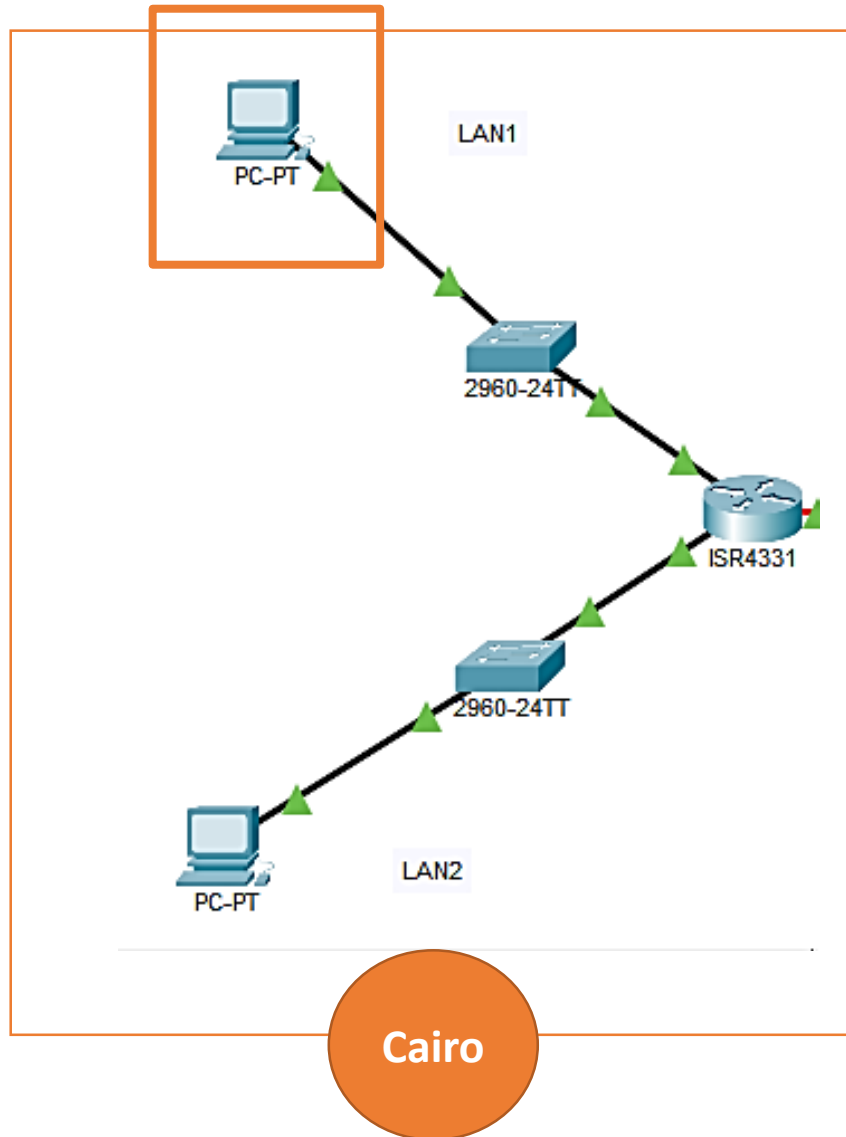
interface g0/0/1
ip address 192.168.1.225 255.255.255.248
no shutdown

interface s0/1/0
ip address 192.168.1.233 255.255.255.252
no shutdown

ip route 192.168.1.128 255.255.255.192 192.168.1.234
ip route 192.168.1.192 255.255.255.224 192.168.1.234

Top
```

PC Configurations



You can add IP and Subnet Mask to each PC, and gateway (router interface IP connected)

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.128

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2D0:BAFF:FE3B:D651

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

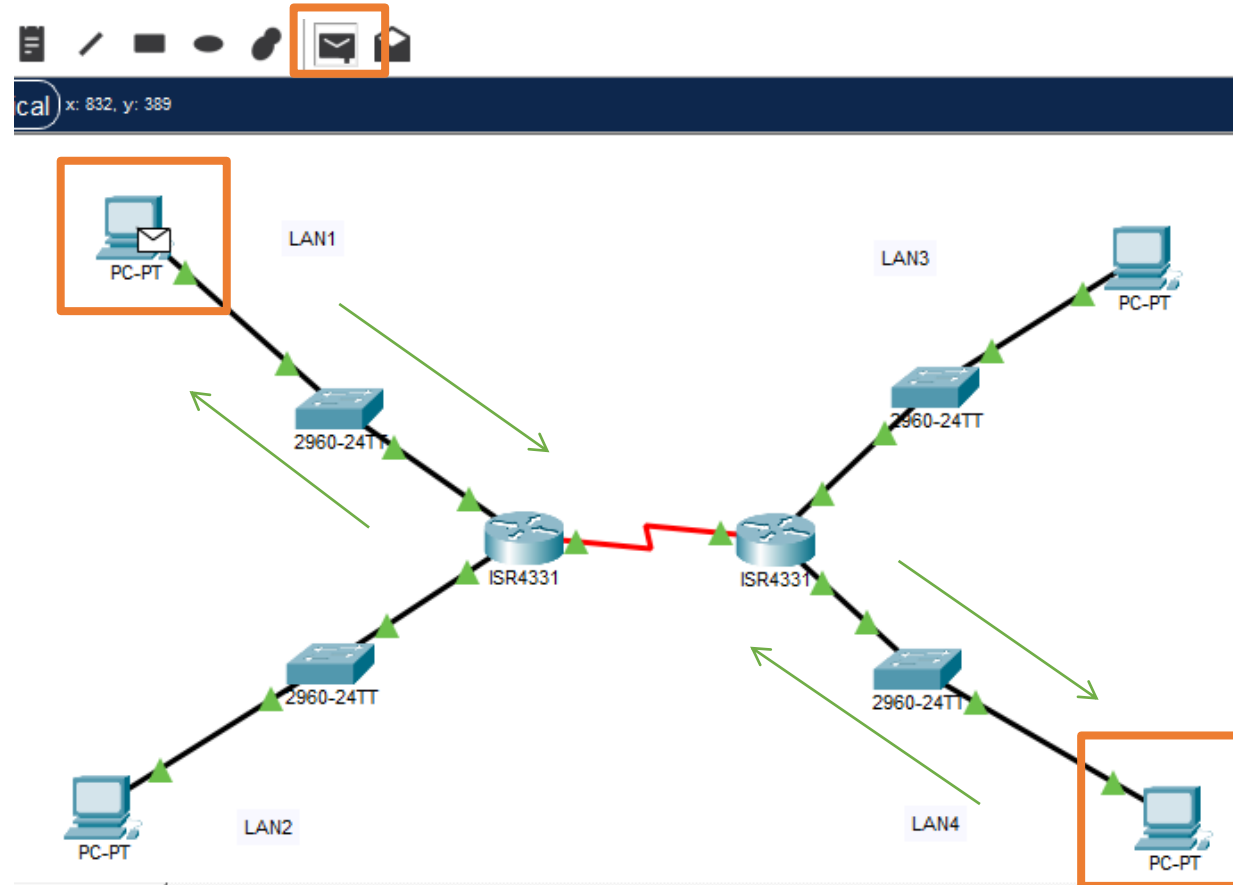
Username

Password

Top

Sending Message

- After inserting all previous steps to all PCs and Routers manually.
- You can send a message from any PC to another by **ping** command in CMD or add simple PDU as figure:



Section 3: DHCP

DHCP

- Dynamic Host Configuration Protocol (DHCP) is a protocol that automatically assigns IP addresses and other network settings to devices in a network.
- Instead of configuring every PC manually (static IP), DHCP gives devices the following automatically when they connect to the network:
 - IP address
 - Subnet mask
 - Default gateway
 - DNS server
- The Dynamic Host Configuration Protocol (DHCP) assigns IP addresses automatically using a 4-step process called DORA.
- This process happens when a device joins the network and needs an IP address.

DHCP

- DORA stands for:
 - D – Discover: PC sends a broadcast message to find a DHCP server.
 - O – Offer: Router suggests an IP address and network configuration
 - R – Request: The client (PC) replies again with Request (accept).
 - A – Acknowledge: It sends ACK containing the full configuration.

In Cisco Packet Tracer when a PC in Cairo LAN1 starts:

- 1) PC → Discover
- 2) Cairo Router → Offer (192.168.1.2)
- 3) PC → Request
- 4) Router → ACK

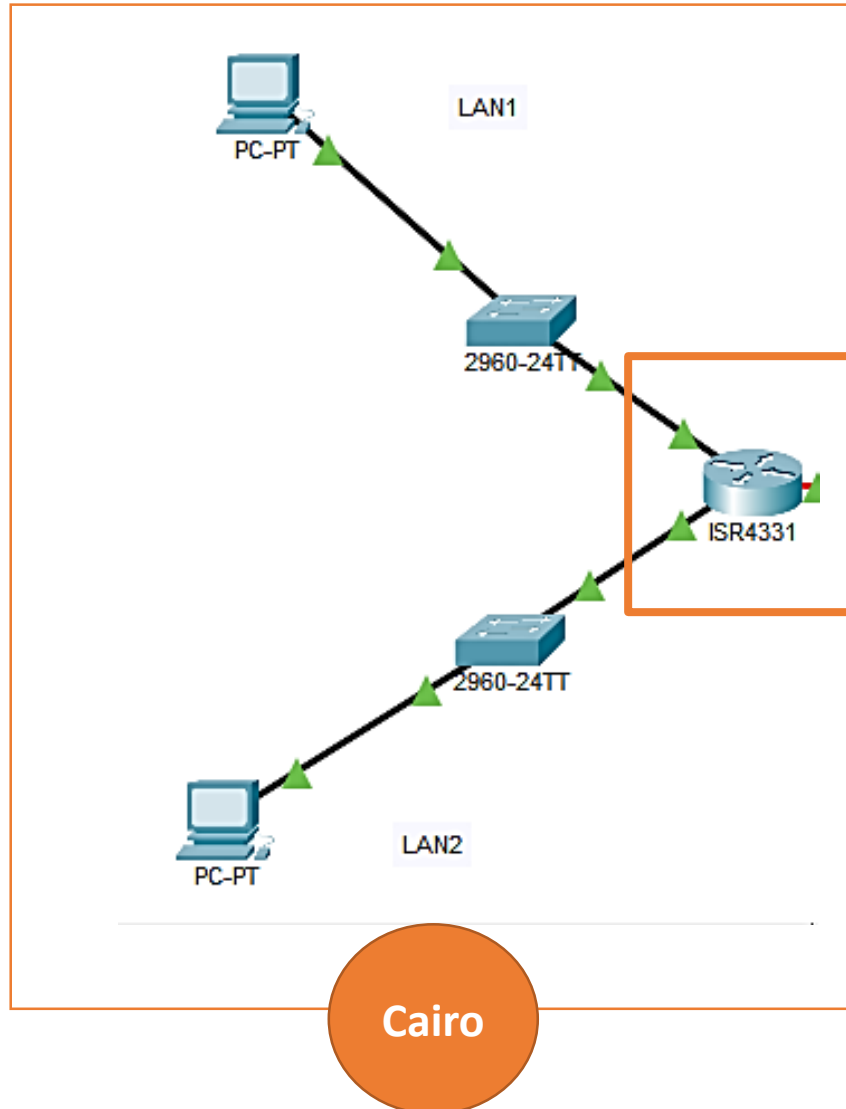
IP: 192.168.1.2
Mask: 255.255.255.128
Gateway: 192.168.1.1

Subnetting using DHCP (Same as Before)

LAN	Network	Mask	Gateway
Cairo LAN1	192.168.1.0/25	255.255.255.128	192.168.1.1
Alex LAN4	192.168.1.128/26	255.255.255.192	192.168.1.129
Alex LAN3	192.168.1.192/27	255.255.255.224	192.168.1.193
Cairo LAN2	192.168.1.224/29	255.255.255.248	192.168.1.225
P2P	192.168.1.232/30	255.255.255.252	routers

Configure Router Interfaces

Add IP and Subnet Mask to each interface and turn on port status.



cairo

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0/0
- GigabitEthernet0/0/1
- GigabitEthernet0/0/2
- Serial0/1/0
- Serial0/1/1

GigabitEthernet0/0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.5818.3101

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.128

Tx Ring Limit 10

Equivalent IOS Commands

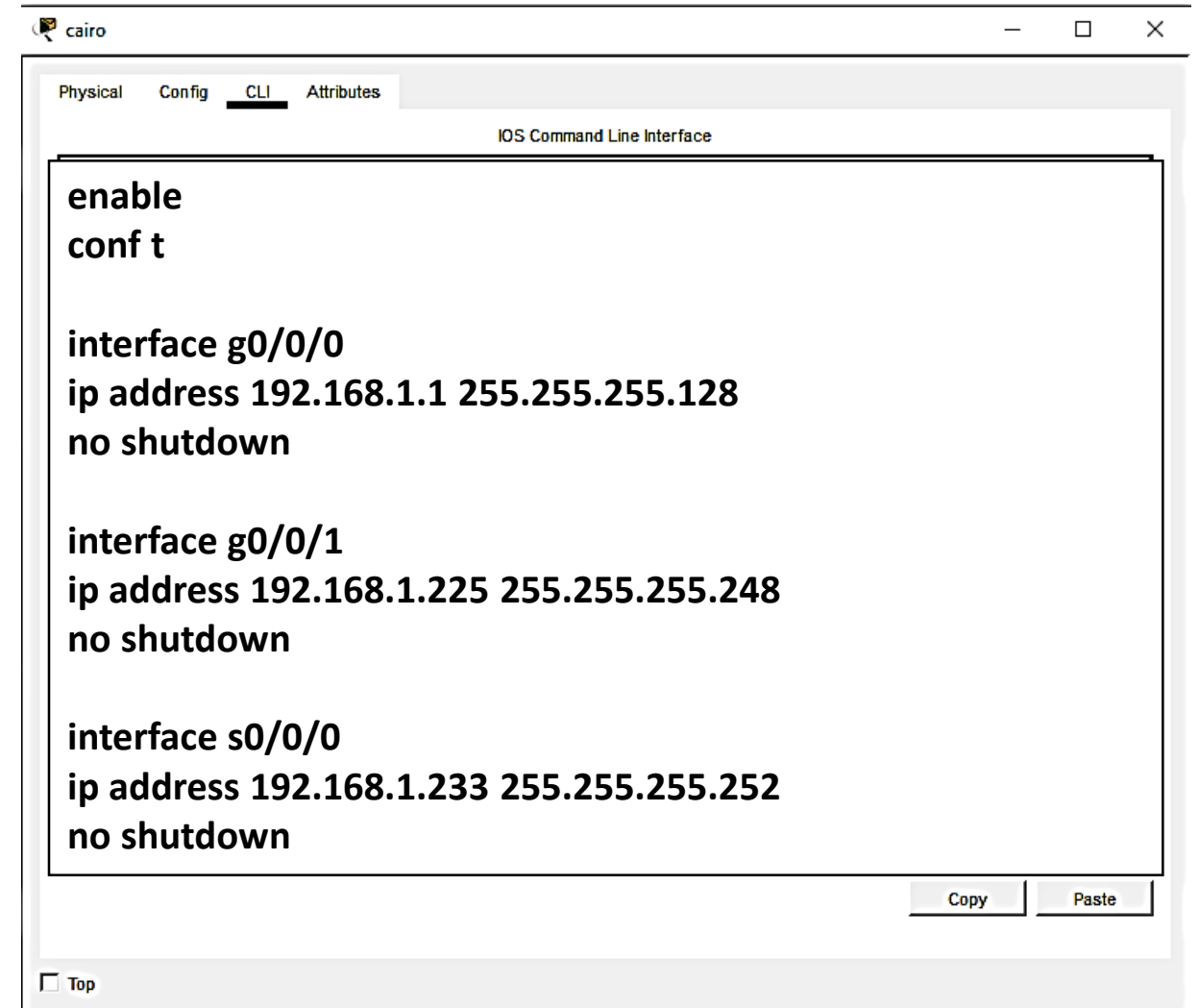
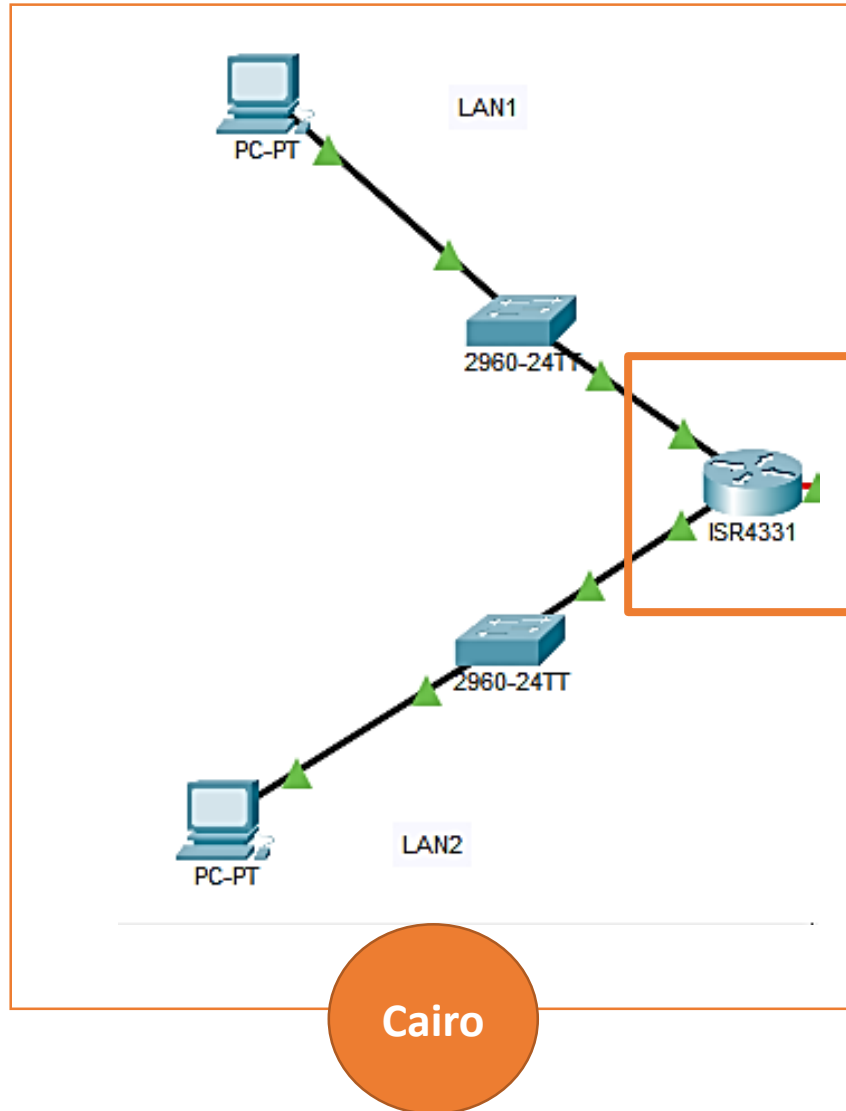
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
```

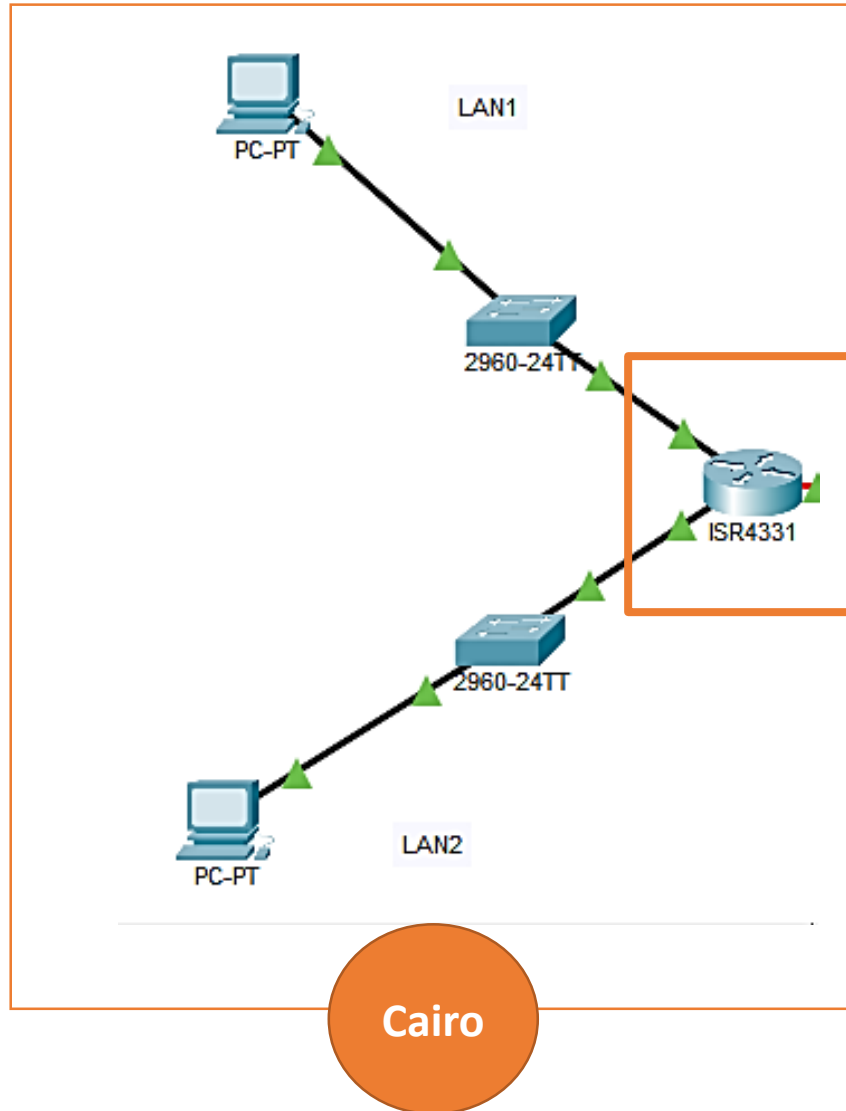
☐ Top

Configure Router Interfaces

You can add IP and Subnet Mask to each interface, turn on port status using CLI commands.



Configure DHCP



```
cairo
Physical Config CLI Attributes
IOS Command Line Interface

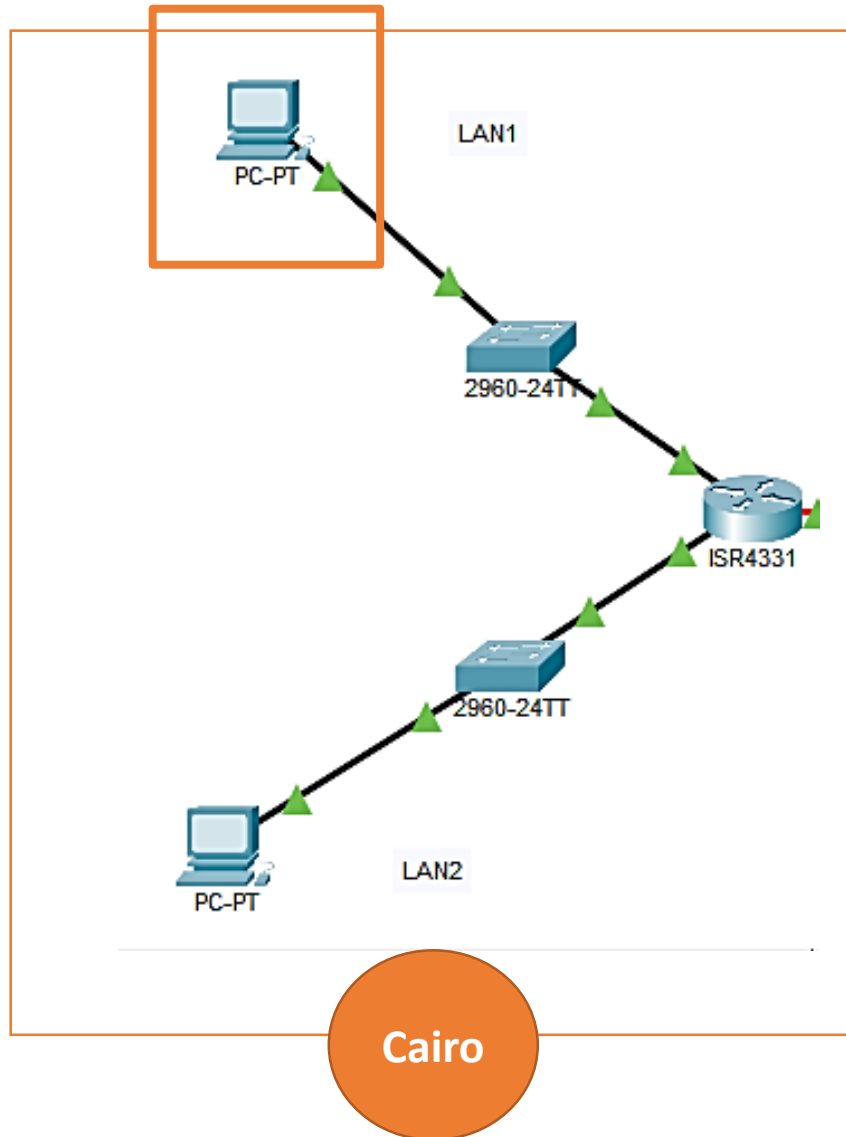
ip dhcp excluded-address 192.168.1.1
ip dhcp excluded-address 192.168.1.225

ip dhcp pool LAN1
network 192.168.1.0 255.255.255.128
default-router 192.168.1.1

ip dhcp pool LAN2
network 192.168.1.224 255.255.255.248
default-router 192.168.1.225

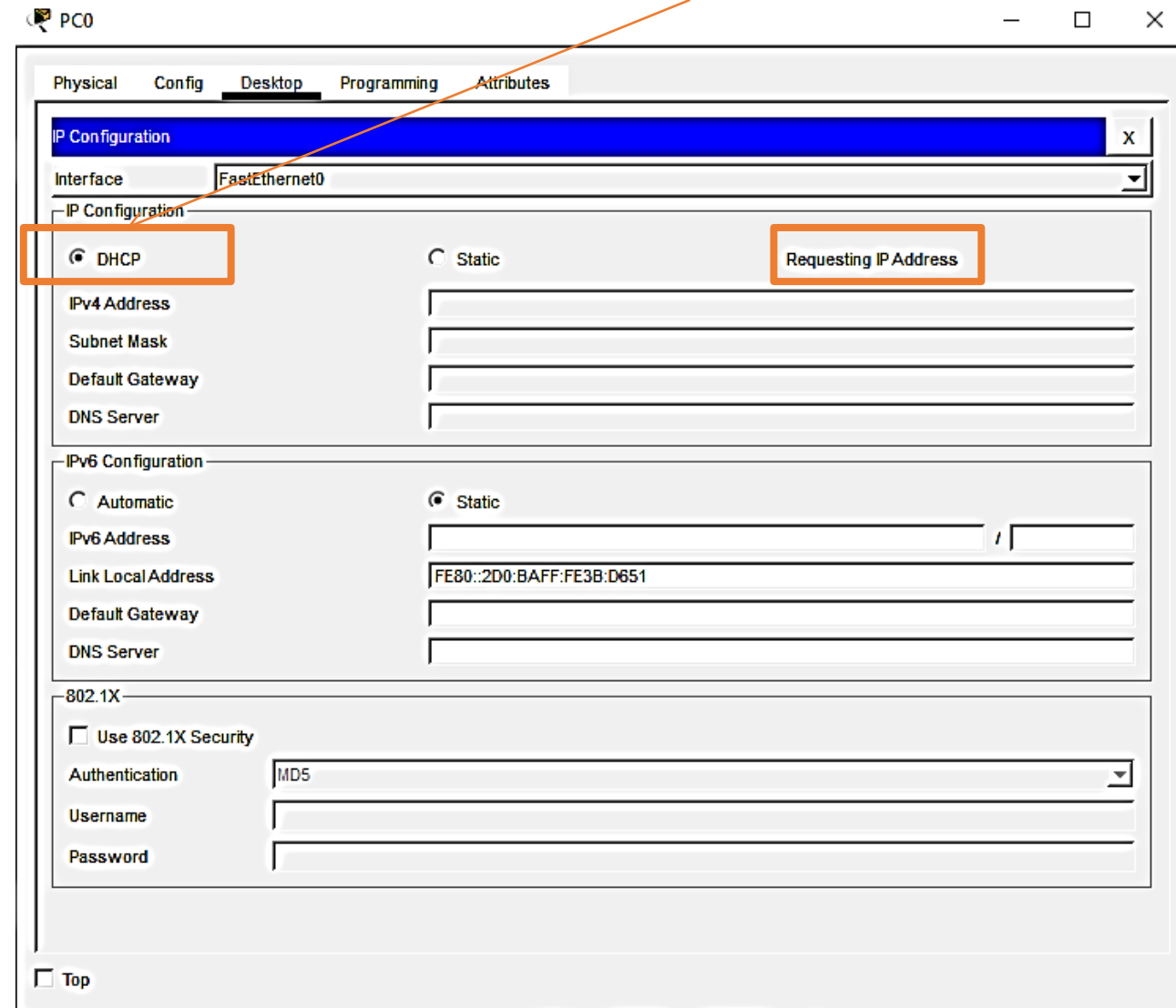
Top
```

Configure PCs

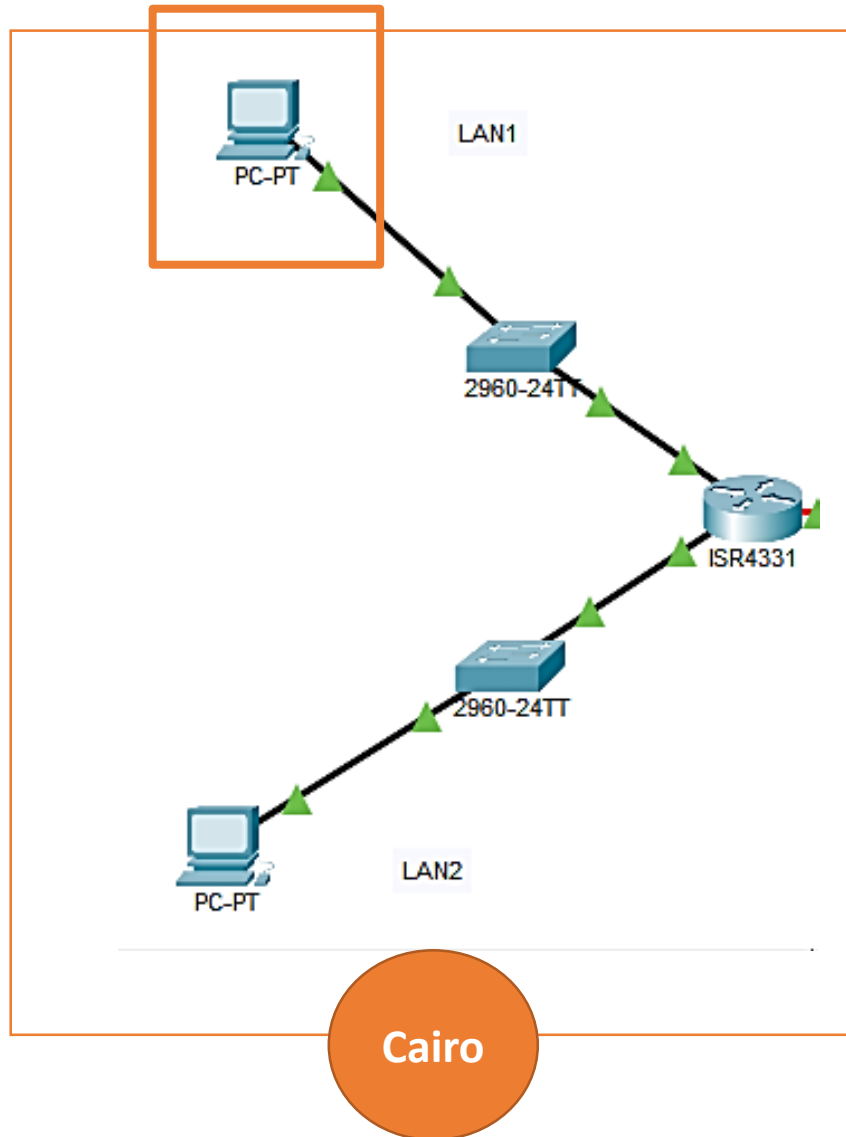


In Cisco Packet Tracer:

- Click the PC
- Go to Desktop
- Open IP Configuration
- Select DHCP



Configure PCs



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static **DHCP request successful.**

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.128

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2D0:BAFF:FE3B:D651

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

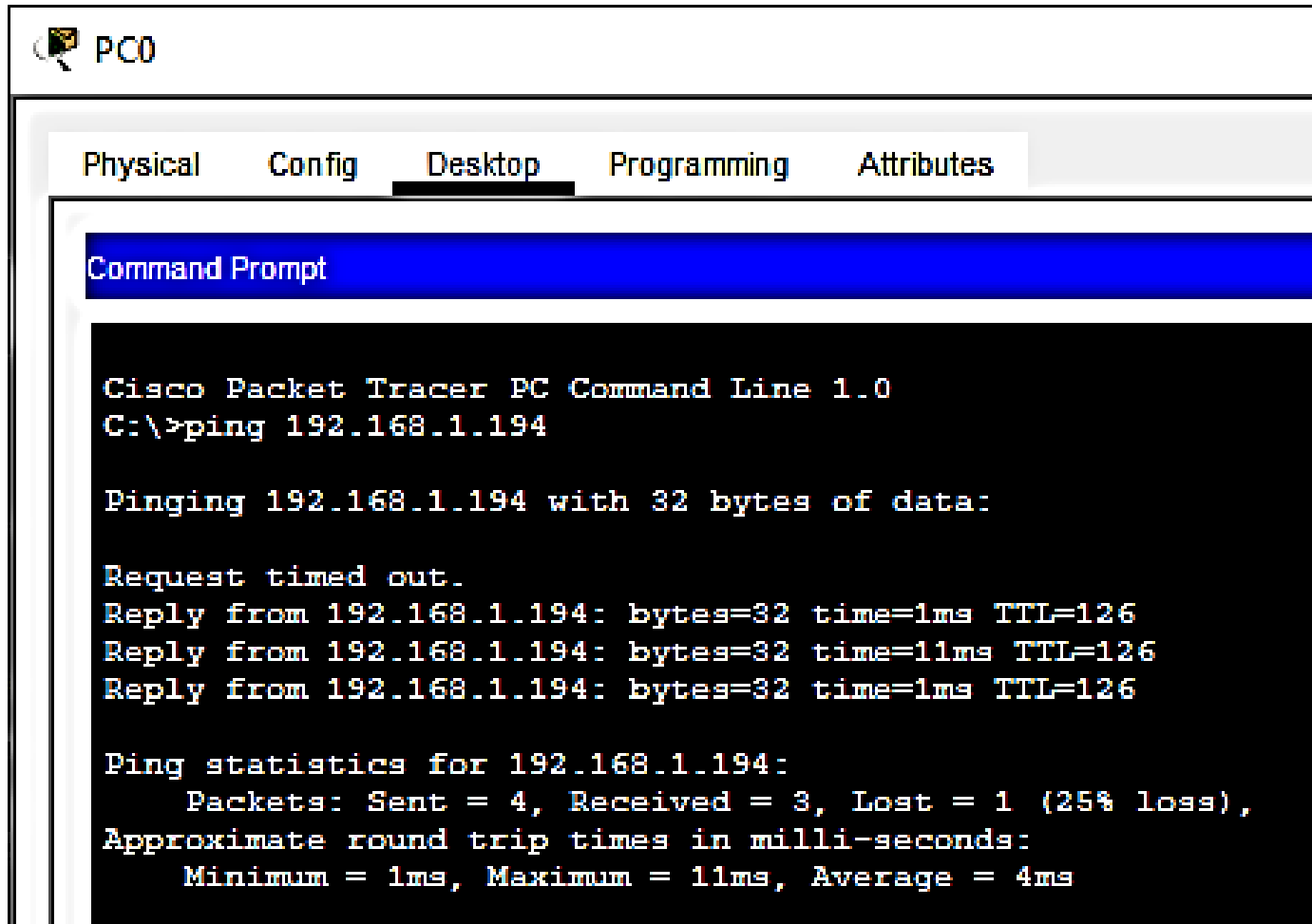
Username

Password

☐ Top

Test Network

- Open command prompt on a PC and test:



The screenshot shows a PC icon labeled 'PC0' with a window titled 'Command Prompt'. The window has tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The command prompt text is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.194

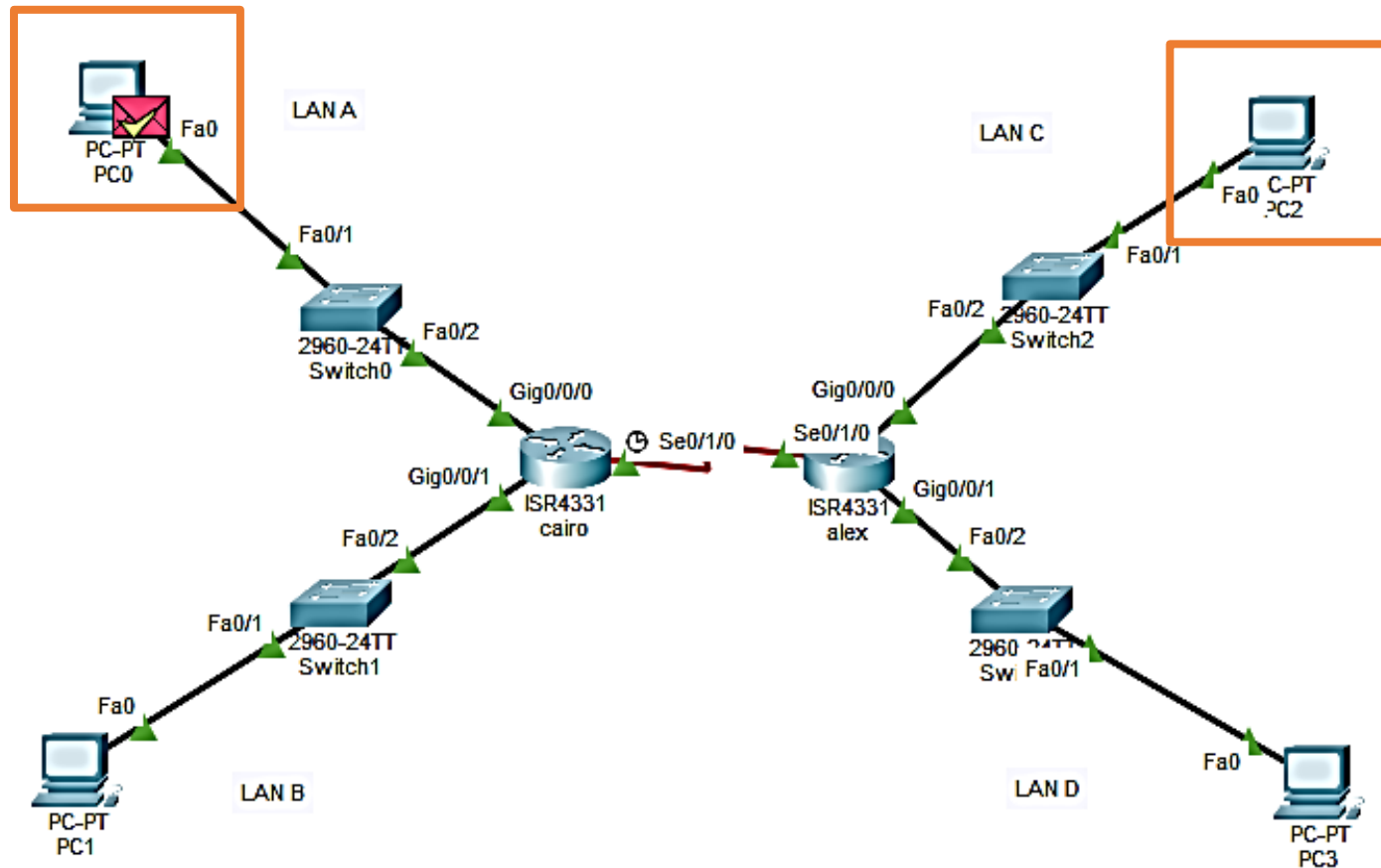
Pinging 192.168.1.194 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.194: bytes=32 time=1ms TTL=126
Reply from 192.168.1.194: bytes=32 time=11ms TTL=126
Reply from 192.168.1.194: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

Sending Message

- After inserting all previous steps to all PCs and Routers using DHCP.
- You can send a message from any PC to another by adding simple PDU as figure:



Simulation Info

The Simulation Panel helps you:

- See how packets travel between devices
- Understand network protocols
- Troubleshoot network problems
- Observe processes like DHCP, ARP, and Ping.

The Simulation Panel interface displays the following components:

- Event List:** A table showing the sequence of events in the simulation.
- Reset Simulation:** A button to reset the simulation.
- Constant Delay:** A checkbox to enable constant delay.
- Captured to:** A label indicating the capture duration (0.010 s).
- Play Controls:** Buttons for play, pause, and stop.
- Event List Filters - Visible Events:** A list of protocols and services currently visible in the event list.
- Edit Filters:** A button to edit the filters.
- Show All/None:** A button to show all or none events.
- Event List, Realtime, Simulation:** A row of buttons to switch between views.
- Table:** A table showing the details of the selected event.

Vis.	Time(sec)	Last Device	At Device
...	0.000	—	PC0
...	0.001	PC0	Switch0
...	0.002	Switch0	cairo
...	0.003	cairo	alex
...	0.004	alex	Switch2
...	0.005	Switch2	PC2
...	0.006	PC2	Switch2
...	0.007	Switch2	alex
...	0.008	alex	cairo

Reset Simulation ☒ Constant Delay Captured to: 0.010 s

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPv6, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	PC0	PC2	ICMP	...	0.000	N	0	(e...	(delete)

Any Questions?

Thanks 😊